



Hierarchical Graph Contrastive Learning for Review-Enhanced Recommendation

Changsheng Shui, Xiang Li, Jianpeng Qi, Guiyuan Jiang, and Yanwei Yu^(✉)

Ocean University of China, Qingdao, China
{scs,lixiang1202}@stu.ouc.edu.cn,
{qijianpeng,jiangguiyuan,yuyanwei}@ouc.edu.cn

Abstract. In comparison to numerical ratings and implicit feedback, textual reviews offer a deeper understanding of user preferences and item attributes. Recent research underscores the potential of reviews in augmenting recommendation capabilities, thereby advancing the deployment of review-enhanced recommendation systems. However, existing methodologies often neglect the significance of rating magnitudes and are susceptible to challenges such as data sparsity and long-tail distribution in real-world contexts. To address these challenges, we propose Hierarchical Graph Contrastive Learning (HGCL) for advancing review-enhanced recommendation systems. HGCL dynamically learns hypergraph structures to capture higher-order correlations among nodes and simultaneously integrates local and global collaborative relations through global-local contrastive learning. Additionally, we propose hierarchical graph contrastive learning methods to better model the intrinsic correlation between ratings and reviews, encompassing aspects such as local-global, cross-rating, and edge-level contrastive learning. Extensive experimentation on five public datasets demonstrates that the proposed method notably outperforms state-of-the-art approaches.

Keywords: Graph Representation Learning · Hypergraph Learning · Contrastive Learning · Recommender Systems

1 Introduction

In recent years, the explosion of information has propelled recommender systems [2, 9, 31, 35], into indispensable tools for e-commerce and social media platforms. Among the core challenges in recommender systems, predicting ratings stands prominently [16, 21, 34, 43]. However, a notable disparity between ratings and implicit feedback (such as purchases, clicks, or favorites) is the concurrent provision of textual reviews by many users. While ratings offer a direct reflection of user preferences for items [10], they often fall short in capturing nuanced sentiments. In contrast, textual reviews encapsulate users' personalized needs and attention to specific aspects, providing essential insights into genuine user

C. Shui and X. Li—Contribute equally to this work.

preferences. The rapid advancement of Natural Language Processing (NLP), has significantly augmented semantic understanding capabilities, such as BERT [11] and its variants [24], and BERT4Rec [25]. These NLP models excel in transforming textual reviews into high-fidelity vector representations, effectively capturing users’ semantic sentiments. Consequently, review-based recommendation systems, empowered by these advancements, have become ubiquitous in recommendation technology landscapes.

Traditional review-based recommendation approaches usually rely on the Bag-of-Words model, neglecting contextual and sequential information within the text, which results in the loss of critical semantic nuances. With the advent of deep learning, various methodologies have integrated review information into graph neural network (GNN) architectures [5, 33, 41]. Notably, GNN-based methods have emerged as potent models for comprehensively understanding user-item interactions. Integrating reviews into GNN-based recommendation models has thus become pivotal in enhancing recommendation effectiveness. Wu et al. [32] propose a novel method named RMG which stands as a pioneering effort, integrating graph signals and review information for recommender systems. Subsequently, Shuai et al. [23] build upon this foundation by leveraging semantic information embedded within review features and then propose their RGCL model. It fine-tunes the influence of neighboring nodes and employs contrastive learning techniques to achieve superior interaction modeling. These GNN-based methods for review-enhance recommendation adeptly model user-item bipartite graphs using graph convolution, seamlessly incorporating review vectors obtained through NLP models as semantic cues for interaction edges. They not only surmount limitations in traditional approaches but also enrich recommender systems with precise semantic understanding, amalgamating interaction details with contextual insights gleaned from reviews.

Despite their efficacy, existing methods grapple with several challenges. *First*, they often rely on stacking multiple layers to capture high-order correlations and constraints, leading to noise accumulation and over-smoothing issues. *Second*, inherent susceptibility to data sparsity and distribution imbalances in user-item interaction data poses a significant hurdle, exacerbated by long-tail distribution patterns and rating imbalances. *Last*, the intrinsic correlation between reviews and ratings remains inadequately addressed, with many recommender systems neglecting rating magnitude and order, contrary to implicit feedback mechanisms. Efforts to mitigate these challenges and further enhance the effectiveness of GNN-based recommender systems are imperative for advancing recommendation technology in evolving information landscapes.

Desired by addressing the aforementioned challenges, we propose a novel **H**ierarchical **G**raph **C**ontrastive **L**earning for review-enhanced recommendation named **HGCL**. Specifically, to mitigate the over-smoothing issue, we design a hypergraph learner to capture high-order dependencies between nodes and employ hierarchical contrastive learning to achieve mutual enhancement between two views. Additionally, we propose multi-rating contrastive learning and global-local contrastive learning to strengthen dependencies between adjacent ratings, and further enhance the intrinsic correlation between ratings and reviews through edge-level contrastive learning.

We highlight the key contributions of this work as follows:

- HGCL explicitly proposes the idea of hierarchical contrastive learning, which not only acquires node features from both local and global views between user-item node pairs but also learns complex interaction information from multi-rating and edge-level perspectives.
- HGCL leverages hypergraph learning to partition the user-item interaction graph into distinct subgraphs, which can dynamically learn hypergraph structures to capture higher-order correlations during training.
- Comprehensive experiments on five real-world datasets are conducted to evaluate model performance, and results show HGCL achieves up to 4.10% and 4.35% performance improvement in terms of MSE and MAE, respectively.

2 Related Work

2.1 GNNs for Review-Enhanced Recommendation

To explicitly leverage collaborative signals within user-item interaction graphs, an increasing number of studies in collaborative filtering (CF) have recently integrated graph convolutional networks (GCNs) to enhance model performance. Traditional GNN-based methods [12, 27, 38] extract information from neighbors in the user-item graph through multi-hop convolutions to exploit high-order correlations defined by initial pairwise links. Although these methods have shown promising results, they overlook the fact that neighbors should be target-relevant. Despite some existing methods utilizing attention mechanisms to assign different weights to neighbors, they encounter two main challenges. Firstly, their performance is hindered by the lack of fine-grained details. Secondly, attention learned solely from interaction data may not accurately capture users' preferences.

To tackle these challenges, review-based recommender systems have been proposed [14, 20, 23, 29]. For instance, RMG [32] is a pioneering effort, integrating graph signals and review information for recommendations. Subsequently, RGCL [23] builds upon this foundation by leveraging semantic information embedded within review features. It fine-tunes the influence of neighboring nodes and employs contrastive learning to achieve superior interaction modeling.

2.2 Contrastive Learning

Contrastive learning has emerged as a powerful self-supervised framework, with its core idea being to promote the closeness of representations among different views of the same object, while simultaneously increasing the dispersion of representations among different objects. The essence of contrastive learning lies in bringing an 'anchor' and a 'positive' sample closer in the embedding space, while simultaneously pushing the anchor away from multiple 'negative' samples [15]. Drawing inspiration from the successes of contrastive learning across various domains [3, 7, 13, 17, 39], many researchers have integrated contrastive

learning into graph data to address the challenges posed by insufficient supervision signals in recommender systems recently [19, 26, 28, 37, 44]. RGCL [23] introduces review-based contrastive learning, where corresponding review representations are selected as positive samples. This innovative approach effectively leverages the semantic richness encapsulated within reviews, contributing to more informed and review-enhanced recommendations. Contrastive learning has emerged as a promising avenue for addressing data sparsity and distribution issues within recommender systems. The adoption of these techniques underscores their potential to enhance recommendation models and enrich the user experience.

3 Problem Definition

We let $\mathbf{U} = \{u_1, u_2, \dots, u_M\}$ denote the set of users and let $\mathbf{V} = \{v_1, v_2, \dots, v_N\}$ denote the set of items, where M and N denote the number of users and items, respectively. Each interaction can be defined as a tuple $(u_i, v_j, y_{u_i v_j}, c_{u_i v_j})$, where $y_{u_i v_j}$ indicates the interaction between u_i and v_j and $c_{u_i v_j}$ is the textual review. Additionally, we utilize a pre-trained BERT-Whitening model to process user u_i 's reviews on item v_j , obtaining the vector $\tilde{c}_{u_i v_j} \in \mathbb{R}^d$ as the review embedding. The collection of all review embeddings can be represented as $\tilde{\mathbf{E}} \in \mathbb{R}^{M \times N \times d}$. We typically use a rating matrix $\mathbf{R}$ to store historical ratings from users to items and $\mathcal{R}$ is the set of all ratings (e.g., $\mathcal{R} = \{1, 2, 3, 4, 5\}$), where $r_{u_i v_j}$ indicates a historical rating from u_i to v_j while $r_{u_i v_j} = 0$ means that v_j is unexposed to u_i .

Here we represent the interaction network as a user-item bipartite graph G , and there are only interactions between users and items. The task of review-enhanced recommendation is to predict missing ratings in the bipartite graph.

4 Methodology

In this section, we provide a comprehensive exposition of the proposed HGCL, as illustrated in Fig. 1. HGCL encompasses five pivotal processes: **(1) Subgraph Partition and Review-aware Graph Convolution**, **(2) Multi-Rating Contrastive Learning**, **(3) Hypergraph Structure Learning**, **(4) Global-Local Contrastive Learning**, and **(5) Edge-Level Contrastive Learning**. Generally, through these five integral components, HGCL stands as a holistic solution designed to enhance the accuracy and efficiency of recommendation, by effectively leveraging collaborative signals, topological features, and high-order correlations across varying ratings within user-item interaction data.

4.1 Subgraph Partition and Review-Aware Graph Convolution

Inspired by previous models [22, 40, 42], we divide the bipartite graph into different rating subgraphs, the initial feature for each node consists of three different indices (identical index, role-aware index, and one-hot index) in each subgraph.

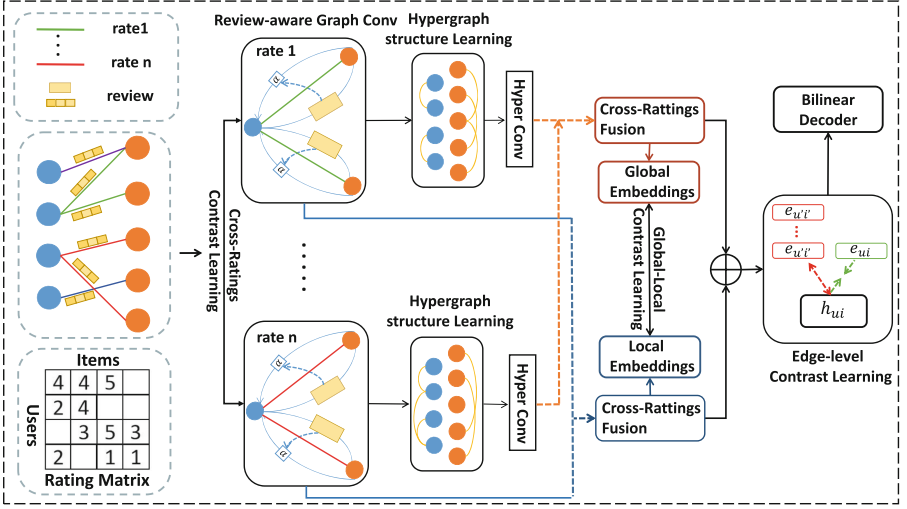


Fig. 1. The overall architecture of the proposed HGCL.

With the feature of each node, we can utilize it to derive the initial embedding of the node from the trainable parameter matrix $E \in \mathbb{R}^{|\mathcal{R}| \times (M+N+3) \times d}$, where $|\mathcal{R}|$ represents the number of rating categories.

As previously mentioned, in contrast to simple rating information, review text provides a more nuanced description of the current interaction edge. It can reflect the user's preference for the item and the popularity of the item itself in greater detail. Therefore, we compute the weight coefficient of different neighbors based on the reviews corresponding to each edge as follows:

$$x_r^{l+1}[u_i] = \sum_{j \in \mathcal{N}_r(u_i)} \frac{\sigma(w_r^l \tilde{c}_{u_i v_j})}{\sqrt{|\mathcal{N}_r(u_i)| \cdot |\mathcal{N}_r(v_j)|}} x_r^l[v_j], \quad (1)$$

where $x_r^{l+1}[u_i]$ denotes the embedding of the u_i at layer $l+1$ in the r -th rating subgraph, $\mathcal{N}_r(u_i)$ is the set of neighbors for u_i in the r -th rating subgraph, w_r^l represents the trainable parameter matrix of the rating subgraph r at the l layer, and $\sigma(\cdot)$ indicates the sigmoid activation function. These embeddings can represent the collaborative signal and the topological information of each user and item on different subgraphs.

4.2 Multi-rating Contrastive Learning

Compared to traditional heterogeneous graphs with multiple edge types, the importance of ratings suggests that users giving nearby ratings share similar preferences, thus leading to similar representations. However, real datasets with imbalanced rating distributions add complexity to this scenario. Partitioning the

heterogeneous graph into rating subgraphs unintentionally results in the loss of sequential rating information.

To tackle this challenge, we propose an innovative multi-rating contrastive learning module that captures the correlation between different ratings by leveraging multi-view contrastive learning. Specifically, we adopt the well-established InfoNCE loss [8] based on multi-rating representations of users and items. We treat a user's embedding learned from the current rating subgraph and the user embedding from the next adjacent rating subgraph as positive pairs while respecting the order of ratings.

$$\mathcal{L}_{MR(r)}^U = \sum_{u \in \mathcal{U}} -\log \frac{\exp((x_u^{(r)} \cdot x_u^{(r+1)} / \tau))}{\sum_{v \in \mathcal{U}} \exp((x_v^{(r)} \cdot x_v^{(r+1)} / \tau))}, \quad (2)$$

where $x_u^{(r)}$ is the normalized output embedding of rating subgraph r and τ is the temperature hyper-parameter of softmax.

As an illustrative example, consider a 5-point Likert scale. In this case, $L_{MR(1)}^U$ denotes the multi-rating contrastive loss between the subgraphs with a rating of '1' and '2'. Similarly, $L_{MR(2)}^U$ signifies the multi-rating contrastive loss between the subgraphs with a rating of '2' and '3', while $L_{MR(3)}^U$ captures the multi-rating contrastive loss between the subgraphs with a rating of '3' and '4'. The overall multi-rating contrastive loss for the entire user can be expressed as follows:

$$\mathcal{L}_{MR}^U = \mathcal{L}_{MR(1)}^U + \mathcal{L}_{MR(2)}^U + \dots + \mathcal{L}_{MR(|\mathcal{R}|-1)}^U. \quad (3)$$

where $|\mathcal{R}|$ denotes the number of rating categories. Similarly, we can obtain the loss function of the item side $\mathcal{L}_{MR}^V$. The complete multi-rating contrastive objective function $\mathcal{L}_{MR}$ is the sum of the above two losses:

$$\mathcal{L}_{MR} = \mathcal{L}_{MR}^U + \mathcal{L}_{MR}^V. \quad (4)$$

4.3 Hypergraph Structure Learning

As previously discussed, the embedding derived from Eq. (2) effectively encapsulates collaborative signals and topological information pertaining to users and items within distinct rating subgraphs. It is worth noting that the inherent topological similarities among nodes often mirror implicit similarities in user preferences or item features. In light of this insight, we endeavor to transcend the conventional limitation inherent in bipartite graphs, where edges typically link nodes of the same type. Drawing inspiration from prior methodologies [31, 35], we employ a multi-layer perceptron (MLP) in the computation of hyperedge assignments for users and items within each rating subgraph. This process yields hypergraphs $\mathbf{H}_u^r$ for users and $\mathbf{H}_v^r$ for items, each reflecting the intricate relations and dependencies that exist within the respective rating contexts:

$$\begin{aligned} \mathbf{H}_u^r &= \text{norm}(\text{LeakyReLU}(x_u^r \mathbf{W}_u^r)), \\ \mathbf{H}_v^r &= \text{norm}(\text{LeakyReLU}(x_v^r \mathbf{W}_v^r)), \end{aligned} \quad (5)$$

where $\mathbf{W}_u^r \in \mathbb{R}^{K \times d}$ and $\mathbf{W}_v^r \in \mathbb{R}^{K \times d}$ are trainable weight matrices, in which K is the number of hyperedges. Upon deriving the user hypergraph and item hypergraph associated with various ratings, we possess the means to incorporate implicit high-order correlations among nodes into the updated representations of both users and items. The hypergraph convolution, which operates independently on each rating subgraph, is formally defined as follows:

$$h_u^r = \mathbf{H}_u^r \mathbf{L}_u^{-1} \mathbf{H}_u^{r\top} x_u^r, \quad h_v^r = \mathbf{H}_v^r \mathbf{H}_v^{-1} \mathbf{H}_v^{r\top} x_v^r. \quad (6)$$

Here, $\mathbf{L}_u$ and $\mathbf{L}_v$ represent the vertex degree matrices associated with the learned user hypergraph $\mathbf{H}_u^r$ and item hypergraph $\mathbf{H}_v^r$, respectively. These matrices play a crucial role in the re-scaling of the resulting embeddings.

4.4 Global-Local Contrastive Learning

As elucidated in Eq. (2) and Eq. (10), each node is endowed with a local embedding capturing the nuances of its local topology, as well as a global embedding encapsulating higher-order dependencies within each rating subgraph. However, in downstream tasks, there often arises a necessity for more concise node embeddings. Conventional approaches typically employ simplistic pooling operations (such as average or sum pooling) on embeddings from various perspectives to derive condensed representations, inadvertently overlooking the importance of node representations pertaining to diverse relations.

To overcome this drawback, we propose a global-local contrastive learning aimed at reinforcing relationships between adjacent ratings. Formally, let $\mathcal{N}_r$ denote the two ratings adjacent to rating r , and let x^r represent the local embeddings of nodes within the r -th rating subgraph. The process of message passing between adjacent ratings unfolds as follows:

$$e^r = x^r + \sum_{r' \in \mathcal{N}_r} \zeta_{r'} \cdot x^{r'}, \quad (7)$$

where $\zeta_{r'}$ is the normalized relevance of rating r' to r , which is calculated by:

$$\zeta_{r'} = \frac{\exp\left(\text{LeakyReLU}\left(q^r x^{r'}\right)\right)}{\sum_{r' \in \mathcal{N}(r)} \exp\left(\text{LeakyReLU}\left(q^r x^{r'}\right)\right)}. \quad (8)$$

q^r constitutes a trainable attention mechanism specifically designed for rating r . It plays a crucial role in governing the flow of information between rating r and its neighboring ratings within $\mathcal{N}(r)$. Similarly, we utilize the global-local contrastive learning in parallel to derive global embeddings γ^r .

Finally, we further process this intermediate output through a linear operator:

$$e[i] = \tanh(\mathbf{W}_1 \sum_{r \in \mathcal{R}} z^r[i]), \quad \gamma[i] = \tanh(\mathbf{W}_2 \sum_{r \in \mathcal{R}} \text{gamma}^r[i]), \quad (9)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ denote the weight parameters of the nonlinear function $\tanh(\cdot)$, $e[\cdot]$ and $\gamma[\cdot]$ represent local and global embeddings after multi-view aggregation, and the final representation of users and items is the combination of local embeddings and global embeddings.

Specifically, we differentiate between the local and global embeddings as two distinct perspectives. In this context, we consider different views of the same user or item as positive pairs, while treating divergent views of users/items as negative pairs. Formally, we introduce our global-local contrastive loss function for user representations, leveraging the InfoNCE framework, as follows:

$$\mathcal{L}_{GL}^U = \sum_{u \in \mathbf{U}} -\log \frac{\exp((e[u] \cdot \gamma[u]/\tau))}{\sum_{u' \in \mathbf{U}} \exp((e[u'] \cdot \gamma[u']/\tau))}. \quad (10)$$

The final global-local contrastive loss is the sum of user and item objectives:

$$\mathcal{L}_{GL} = \mathcal{L}_{GL}^U + \mathcal{L}_{GL}^V. \quad (11)$$

4.5 Edge-Level Contrastive Learning

The preceding multi-rating and local-global methods represent variants of multi-view contrastive learning. These methodologies aim to foster the congruence of identical nodes across diverse views while emphasizing the divergence of distinct nodes, thereby facilitating synergistic augmentation and information complementarity across varied perspectives. Nevertheless, while embeddings predominantly capture the characteristics of associated interaction edges, the aforementioned contrastive learning techniques fail to account for the self-supervised cues inherent within interaction edges and their corresponding reviews. Drawing inspiration from prior methodologies, we capitalize on the richness of review data by employing edge-level contrastive learning to fully harness the potential of review information.

To capture the interaction edge between users and items, we employ a Multi-layer Perceptron (MLP) to construct the interaction embedding $h_{u_i v_j}$. This embedding is derived by concatenating the final representations of users and items.

$$h_{u_i v_j} = \text{MLP}(e[u_i], e[v_j]), \quad (12)$$

where $e[u_i]$ and $e[v_j]$ represent the final embeddings of user u_i and item v_j respectively, and MLP comprises two hidden layers with a ReLU activation function. Subsequently, all interaction embeddings serve as anchor examples, while the corresponding review embeddings are designated as positive samples. However, in the case of an interaction embedding with an actual rating of ‘1’, for instance, the ideal negative sample should be chosen from the review embeddings with a rating of ‘5’. Nevertheless, RGCL (presumably a previous method) imposes no constraints on the random sampling range of negative samples, and each anchor of RGCL corresponds to only one negative sample, thereby diminishing the effectiveness of contrastive learning. Hence, HGCL takes into account the sequentiality of ratings and the distribution of the number of interactions

corresponding to different ratings and introduces a method to extract negative samples based on ratings. In essence, each rating selectively chooses negative samples from reviews associated with ratings that are farther away. The edge-level contrastive learning mechanism can be formulated as follows:

$$\mathcal{L}_{EL} = \sum_{u_i, v_j \in \mathcal{E}} -\log \frac{\exp((h_{u_i v_j} \cdot \tilde{c}_{u_i v_j} / \tau))}{\sum_{c'_{u_i v_j} \in \mathcal{N}(h_{u_i v_j})} \exp((h_{u_i v_j} \cdot c'_{u_i v_j} / \tau))}. \quad (13)$$

4.6 Model Optimization

In line with prior work [1, 22], we adopt a bilinear decoder for the purpose of reconstructing edges within the user-item graph. The decoder functions by bilinearizing the potential rating levels, generating a probability distribution, and subsequently applying a softmax function:

$$p(\check{\mathbf{R}}_{u_i v_j} = r) = \frac{e^{(e[u_i])^T Q_r e[v_j]}}{\sum_{r'=1}^{\mathcal{R}} e^{(e[u_i])^T Q_{r'} e[v_j]}}, \quad (14)$$

where Q_r is the learnable parameter matrix of rating r . The predicted rating matrix is calculated as:

$$\check{\mathbf{R}}_{u_i v_j} = \sum_{r \in \mathcal{R}} r p(\check{\mathbf{R}}_{u_i v_j} = r). \quad (15)$$

We minimize the cross-entropy loss (denoted by CE) between the predictions and the ground truth ratings. While assigning a value of ‘1’ to each ground truth label, we also set two adjacent labels to a smaller value, denoted as l_{close} , mitigating the problem that CE loss ignores differences between incorrect labels. Typically, we set l_{close} to 0.1 to strike an appropriate balance.

$$\mathcal{L}_{main} = \frac{1}{|(u_i, v_j) | \Omega_{u_i, v_j} = 1|} \sum_{(u_i, v_j) | \Omega_{u_i, v_j} = 1} \mathbf{CE}(r[u_i, v_j], \hat{r}[u_i, v_j]), \quad (16)$$

where we represent the expanded true rating label as $r[u_i, v_j]$, while $\hat{r}[u_i, v_j]$ signifies the predicted rating for the pair (i, j) . In parallel, we introduce hyper-parameters α and β to weigh and optimize the contrastive learning tasks. This leads us to the final loss function, which is defined as follows:

$$\mathcal{L} = \mathcal{L}_{main} + \alpha \mathcal{L}_{MR} + \beta \mathcal{L}_{GL} + \theta \mathcal{L}_{EL} + \varphi \mathcal{L}_{NRR}. \quad (17)$$

Here, we introduce $\varphi \mathcal{L}_{NRR}$ as a regularization mechanism inspired by prior models [1, 22]. This regularization method is designed to account for both magnitude and order information, which aims to foster similarity in the representation of each node in the rating subgraph that is adjacent to each other.

5 Experiment

5.1 Experimental Settings

Datasets. We use two publicly benchmark data source, Amazon and Yelp, where Amazon includes “Digital_Music”, “Toys_and_Games”, “Clothing” and “CDs_and_Vinly” four domains. Following the preprocessing steps outlined in [23], we convert all datasets to “5-core” format, which means there are at least five reviews for each user or item. Table 1 presents the summary of dataset statistics.

Table 1. Statistical summaries of datasets, U: number of users; I: number of items; R: rating counts; We present percentage distributions of user ratings.

Dataset	U	I	R	density(%)	Ratings%				
					1	2	3	4	5
Yelp	8,423	3,742	88,647	0.281	5.88	9.41	15.29	37.13	32.09
Digital_Music	5,541	3,568	64,706	0.330	4.31	4.65	10.49	25.55	54.98
Clothing	39,387	23,033	278,677	4.01	5.55	10.91	27.14	20.94	58.57
Toys_and_Games	19,412	11,924	167,597	0.072	2.80	3.75	9.75	22.34	61.33
CDs_and_Vinly	75,258	64,443	1,097,592	0.023	4.06	4.06	8.86	21.47	57.06

Evaluation Protocols and Metrics. We split our training, validation, and test datasets in the ratio of 8:1:1 following the same settings in the most recent studies [22, 40]. In order to mitigate the sampling bias and to fairly evaluate the prediction results of the ratings, we set up MAE and MSE, two widely used metrics in the field of matrix complementation.

Baselines. The baselines can be broadly classified into three distinct groups: (1) **Traditional methods:** SVD [18] is a matrix factorization model that uses the inner product of users’ and items’ latent factors to estimate ratings. NNMF [4] is a classical matrix factorization model that uses the user-item ratings only as the target value of their objective function. (2) **Review-based methods:** DeepCoNN [1] is one of the pioneer works that applies a neural network to a review recommendation system, using two parallel CNNs to capture the latent factors of users and items. NARRE [22] is a model based on the review level attention mechanism, which estimates the importance of different reviews. DAML [36] uses both local and interactive attention layers to learn the embedding of users and items when doing CNN convolution. TransNets [42] converts the learned information into an approximation of the corresponding review through a transformation layer, and uses it for rating prediction. (3) **Graph and review fusion**

methods: RMG [32] fuses review information in graph structure to select important words, sentences, and comments through a three-level attention network. SSG [6] augments the existing single-view approach by merging two additional views and constructs a three-way encoder to capture the long-term, short-term, and collaborative features of users and items for recommendation. RGCL [23] is a graph-based contrastive learning framework that creates a review-aware user-item graph with edge features enriched by review content and introduces two additional contrastive learning tasks to provide self-supervised signals.

Parameter Setting. We employ the Adam optimizer for the entirety of model training, initializing all trainable parameters using the Xavier method. We systematically explore several key hyperparameters: The number of layers is varied within the range $\{1, 2, \dots, 5\}$. The hyperparameters α , β and θ in a range of $\{0.001, 0.01, 0.02, 0.05, 0.1, 0.2\}$, λ is searched in $\{1e^{-7}, 1e^{-6}, 1e^{-5}, 1e^{-4}\}$. In the embedding layer, we consider different vector sizes, selecting from 30, 60, ..., 1200, and the dimension of the review embedding generated by BERT-Whitening is set to 64. The number of hyperedges is searched in $\{4, 8, 16, 32, 128\}$. Additionally, we implement early stopping to mitigate the risk of overfitting. The complete model implementation is carried out using the Deep Graph Library [30] and PyTorch, harnessing the computational power of an Intel Xeon Gold 5320T CPU (2.3GHz) with two NVIDIA 3090 GPUs. The source code of our model is available at: <https://github.com/lx970414/PKDD24/tree/master>.

5.2 Comparison and Result Analysis

Performance comparison of baselines and our proposed HGCL has been shown in the Table 2. Compared with traditional recommendation models based matrix factorization, SVD and NMF, other GNN-based baselines that fusing graph structure information have achieved better performance, which proves the advantages of graph learning for modeling the interaction between nodes.

Specifically, for review-based methods, Deep- CoNN and DAML both use CNNs to capture the latent factors of users and items, which performs well but ignores important graph-level information. NARRE is a review-based model with a review-level attention mechanism, which estimates the importance of different reviews, but it neglects high-order information of nodes. TransNets converts the learned information into an approximation of the corresponding review through a transformation layer. However, it brings a lot of extra time overhead and the experimental performance is inferior to our proposed HGCL.

For the graph and review fusion method, RMG, uses review information in graph structure to select important context through a three-level attention network, but it only focuses on local information of nodes and reviews which leads to the lack of high-order information. For SSG and RGCL, which also incorporate graph structure information, perform worse than the other two fusion methods. This can be attributed to the fact that SSG has difficulty in to adequately capturing collaborative information shared between users and items. Furthermore,

HGCL achieves the best performance across all datasets and shows a significant improvement compared to the latest graph-review fusion method RGCL. This can be attributed to hierarchical graph contrastive learnings of HGCL, which alleviative long-tail distribution by rendering the learned representation distribution more uniform. Additionally, HGCL captures high-order correlations between nodes through the hypergraph learner and uses cross-view contrastive learning to achieve mutual enhancement of information between different views. Finally, HGCL also captures the intrinsic correlation between ratings and reviews through edge-level contrastive learning.

Table 2. Performance comparison of the proposed model and different baselines.

Method	Digital_Music		Toys_and_Games		Clothing		CDs_and_Vinly		Yelp	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
SVD	0.8523	0.6879	0.8301	0.6812	1.1167	0.8082	0.8662	0.8246	1.1741	0.8705
NNMF	0.8316	0.6960	0.8296	0.6822	1.1156	0.8095	0.8663	0.8257	1.1695	0.8637
DeepCoNN	0.8378	0.7032	0.8207	0.6739	1.1184	0.8113	0.8621	0.8231	1.1671	0.8611
NARRE	0.8172	0.6616	0.8175	0.6730	1.1064	0.8077	0.8495	0.8065	1.1673	0.8551
DAML	0.8237	0.6597	0.8143	0.6714	1.1065	0.8082	0.8483	0.8072	1.1588	0.8471
TransNets	0.8273	0.6788	0.8163	0.6782	1.1141	0.8127	0.8440	0.7986	1.1661	0.8447
RMG	0.8074	0.6635	0.8092	0.6586	1.1064	0.8067	0.8425	0.7991	1.1507	0.8433
SSG	0.8218	0.6780	0.8194	0.6770	1.1228	0.8135	0.8458	0.8013	1.1613	0.8529
RGCL	<u>0.7735</u>	<u>0.6524</u>	<u>0.7984</u>	<u>0.6502</u>	<u>1.0858</u>	<u>0.7783</u>	<u>0.8180</u>	<u>0.7729</u>	<u>1.1397</u>	<u>0.8394</u>
HGCL	0.7585	0.6240	0.7657	0.6355	1.0753	0.7731	0.8059	0.7624	1.1172	0.8097

5.3 Ablation Study

To evaluate the effectiveness of each component, we further conduct the ablation study on different HGCL variations. We report the results of the ablation study on five datasets in Table 3. Specifically, we generate the following variants:

- **w/o R** - which removes review-aware graph convolution.
- **w/o CLH** - which removes all the hierarchical contrastive learnings and hypergraph convolution.
- **w/o M-R** - which removes the multi-ratings contrastive learning.
- **w/o G** - which removes the hypergraph structure learning module and hypergraph convolution on learned hypergraph, so that the final representation of the node contains only local embeddings.
- **w/o ECL** - which removes edge-level contrastive learning.

Based on the results, we have the following four observations: (1) The results of **w/o ECL** and **w/o R** demonstrate the necessity of considering the intrinsic correlation between ratings and reviews in HGCL. Furthermore, **w/o CLH** performs worse than all variants on all datasets, which is equivalent to degenerating into a normal GAE-based model. (2) Both **w/o M-R** and **w/o G** have decreased compared to HGCL, which indicates the effectiveness of these two

Table 3. Performance comparison of different variants (MSE)

Dataset	Digital_Music	Toys_and_Games	Clothing	CDs_and_Vinly	Yelp
w/o R	0.7638	0.7731	1.0829	0.8133	1.1229
w/o CLH	0.7922	0.7989	1.1124	0.8122	1.1381
w/o M-R	0.7809	0.7863	1.0848	0.8225	1.1204
w/o G	0.7694	0.7742	1.0913	0.8159	1.1259
w/o ECL	0.7642	0.7701	1.0811	0.8127	1.1241
HGCL	0.7585	0.7657	1.0753	0.8059	1.1172

modules. Specifically, we can see that the two modules contribute differently to different datasets, which may be affected by the situation of specific datasets. For example, multi-ratings contrastive learning has significantly improved performance in Toys and CDs, which may be due to the most prominent issue of rating imbalanced distribution. The better performance of hypergraph learning for Yelp and Clothing may be due to the higher density of these two datasets and more complex dependency relations between nodes, thus requiring hypergraphs to represent the global collaborative information.

5.4 Impact of Imbalanced Data Distribution

To validate HGCL’s efficacy in addressing popularity bias, we conduct experiments assessing the impact of imbalanced node interaction distribution and rating distribution. In the interest of brevity, we present the results specifically for Digital_Music datasets. However, it is worth noting that similar observations are made across other datasets, reinforcing the robustness of our findings.

Impact of Imbalanced Node Interactions. In this section, we assess the performance of HGCL across users exhibiting varying degrees of sparsity in their interactions. To achieve this, we categorize all users into three distinct subsets, based on their interaction frequencies. Specifically, we designate 80% of users with the fewest interactions as ‘Inactive’, classify the top 5% of users with the most interactions as ‘Active’, and assign the remaining users as ‘Normal’. The resulting finding is visually depicted in Fig. 2(a). Notably, this figure reveals a consistent trend wherein HGCL consistently outperforms the two baseline models. Intriguingly, the observed improvements are most pronounced among users with lower interaction frequencies. This phenomenon can be attributed to the effectiveness of InfoNCE losses, which address the popularity bias issue by promoting uniformity in the learned representations.

Impact of Imbalanced Rating Interactions. In this section, we present an analysis of HGCL’s performance across different rating categories. The outcomes of this evaluation are visually depicted in Fig. 2(b). Our findings underscore the significant improvements achieved, particularly in the context of long-tail ratings

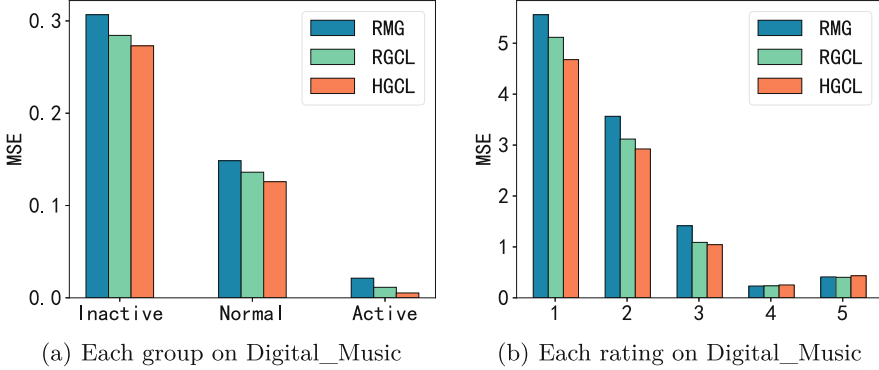


Fig. 2. Impact of imbalanced data distribution.

(‘1’ and ‘2’). Importantly, these enhancements do not come at the expense of performance in more common data categories (head class). This compelling result reaffirms the role of InfoNCE in reshaping the overall distribution of embeddings, aligning them more closely with the characteristics of long-tail rating distributions.

5.5 Parameter Sensitivity

In our evaluation, we systematically investigate the sensitivity of HGCL with regard to four primary hyperparameters: α , β , θ , and the number of hyperedges. The corresponding MSE results obtained under different hyperparameter settings are presented Fig.3. For the sake of conciseness, we present results exclusively for the ML-100K and Amazon datasets; however, it’s important to emphasize that our observations hold true across other datasets as well.

Effect of α , β and θ . To assess the impact of three contrastive learning losses, we conduct a series of parameter sensitivity experiments. The range for α , β and θ is varied from 0.001 to 0.2. The resulting insights of three hyperparameters are presented in Fig.3(a) and 3(b), respectively. We observe similar trends in the variation of three parameters across the two datasets. Initially, as the values increase, the performance of HGCL improves on both datasets. This indicates that contrastive learning effectively facilitates coordinated node representation learning within HGCL. However, it is noteworthy that when the parameters become too large, they might affect the importance of the loss function of the specific task in the model learning, thus damaging the model performance.

Effect of Hyperedge Number. This part delves into the exploration of the optimal number of hyperedges. We conduct experiments by varying the hyperedge numbers, ranging from 4 to 128 on two datasets. The resulting insights are presented in Fig.3(c) and 3(d). For Digital Music, the optimal number of

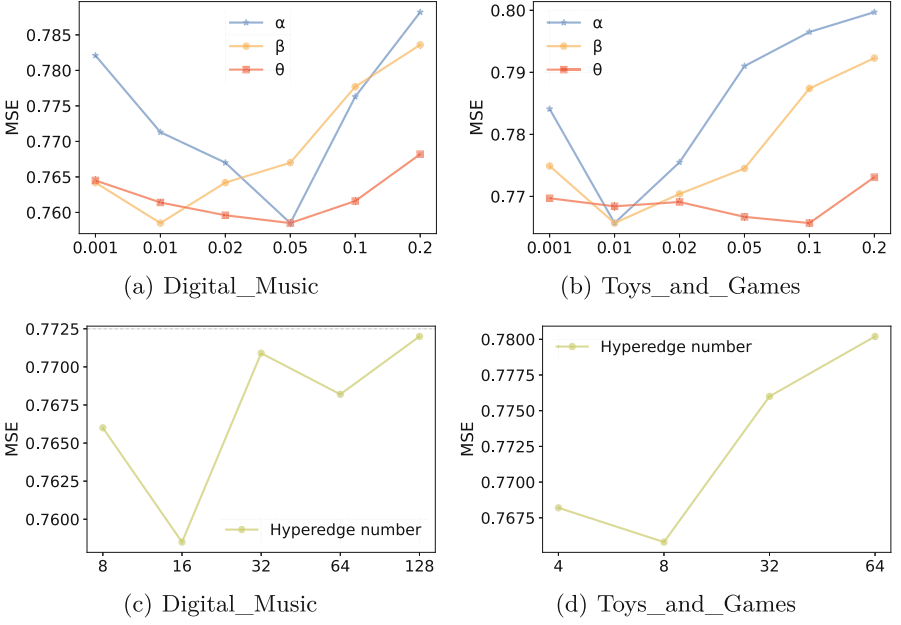


Fig. 3. The parameter sensitivity results.

hyperedges appears to be 16, while on Toys and Games, the model’s performance reaches its peak at 8. Interestingly, further increasing the number of hyperedges beyond these optimal values tends to degrade model performance to varying degrees, which can be attributed to the fact that an excessive number of hyperedges can introduce unnecessary noise into the representations.

6 Conclusion

In this paper, we introduce a novel Hierarchical Graph Contrastive Learning (HGCL) for review-enhanced recommendation. First, we apply review-aware graph convolution on each rating subgraph, dynamically learning the hypergraph structure of different subgraphs. Second, we employ a cross-rating contrastive learning to enhance mutual reinforcement by encouraging alignment between adjacent views. Third, we use global-local contrastive learning to collaboratively learn node representations. Additionally, we design an edge-level contrastive learning to strengthen the intrinsic correlation between reviews and ratings. Experiment results on five benchmark datasets show the superiority and effectiveness of the proposed HGCL.

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References

1. Berg, R.V.D., Kipf, T.N., Welling, M.: Graph convolutional matrix completion. arXiv preprint [arXiv:1706.02263](https://arxiv.org/abs/1706.02263) (2017)
2. Cai, X., Huang, C., Xia, L., Ren, X.: Lightgcl: simple yet effective graph contrastive learning for recommendation. In: ICLR (2022)
3. Chen, T., Kornblith, S., Norouzi, M., Hinton, G.: A simple framework for contrastive learning of visual representations. In: ICML, pp. 1597–1607. PMLR (2020)
4. Dziugaite, G.K., Roy, D.M.: Neural network matrix factorization. arXiv preprint [arXiv:1511.06443](https://arxiv.org/abs/1511.06443) (2015)
5. Fu, C., Zheng, G., Huang, C., Yu, Y., Dong, J.: Multiplex heterogeneous graph neural network with behavior pattern modeling. In: SIGKDD, pp. 482–494 (2023)
6. Gao, J., Lin, Y., Wang, Y., Wang, X., Yang, Z., He, Y., Chu, X.: Set-sequence-graph: a multi-view approach towards exploiting reviews for recommendation. In: CIKM, pp. 395–404 (2020)
7. Giorgi, J., Nitski, O., Wang, B., Bader, G.: Declutr: deep contrastive learning for unsupervised textual representations. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), pp. 879–895 (2021)
8. Gutmann, M., Hyvärinen, A.: Noise-contrastive estimation: a new estimation principle for unnormalized statistical models. In: ICAIS, pp. 297–304. JMLR Workshop and Conference Proceedings (2010)
9. Ji, S., Feng, Y., Ji, R., Zhao, X., Tang, W., Gao, Y.: Dual channel hypergraph collaborative filtering. In: SIGKDD, pp. 2020–2029 (2020)
10. Kalofolias, V., Bresson, X., Bronstein, M., Vandergheynst, P.: Matrix completion on graphs. In: Neural Information Processing Systems Workshop (2014)
11. Kenton, J.D.M.W.C., Toutanova, L.K.: Bert: pre-training of deep bidirectional transformers for language understanding. In: NAACL-HLT, pp. 4171–4186 (2019)
12. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: ICLR (2016)
13. Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., Soricut, R.: Albert: a lite BERT for self-supervised learning of language representations. In: ICLR (2019)
14. Liu, M., Xu, X., Li, J., Li, G.: A review-based feature-level information aggregation model for graph collaborative filtering. Neurocomputing 126697 (2023)
15. Ma, Y., He, Y., Zhang, A., Wang, X., Chua, T.S.: Crosscbr: cross-view contrastive learning for bundle recommendation. In: SIGKDD, pp. 1233–1241 (2022)
16. Mahadevan, A., Arock, M.: A class imbalance-aware review rating prediction using hybrid sampling and ensemble learning. Multimedia Tools Appl. **80**, 6911–6938 (2021)
17. Oord, A.v.d., Li, Y., Vinyals, O.: Representation learning with contrastive predictive coding. arXiv preprint [arXiv:1807.03748](https://arxiv.org/abs/1807.03748) (2018)

18. Rao, N., Yu, H.F., Ravikumar, P.K., Dhillon, I.S.: Collaborative filtering with graph information: consistency and scalable methods. In: *Advances in Neural Information Processing Systems*, vol. 28 (2015)
19. Ren, Y., Liu, B.: Heterogeneous deep graph infomax. In: *Workshop of Deep Learning on Graphs: Methodologies and Applications co-located with the Thirty-Fourth AAAI Conference on Artificial Intelligence* (2020)
20. Ren, Y., et al.: Disentangled graph contrastive learning for review-based recommendation. *arXiv preprint [arXiv:2209.01524](https://arxiv.org/abs/2209.01524)* (2022)
21. Seo, S., Huang, J., Yang, H., Liu, Y.: Interpretable convolutional neural networks with dual local and global attention for review rating prediction. In: *RecSys*, pp. 297–305 (2017)
22. Shen, W., Zhang, C., Tian, Y., Zeng, L., He, X., Dou, W., Xu, X.: Inductive matrix completion using graph autoencoder. In: *CIKM*, pp. 1609–1618 (2021)
23. Shuai, J., et al.: A review-aware graph contrastive learning framework for recommendation. In: *SIGIR*, pp. 1283–1293 (2022)
24. Su, J., Cao, J., Liu, W., Ou, Y.: Whitening sentence representations for better semantics and faster retrieval. *arXiv preprint [arXiv:2103.15316](https://arxiv.org/abs/2103.15316)* (2021)
25. Sun, F., Liu, J., Wu, J., Pei, C., Lin, X., Ou, W., Jiang, P.: Bert4rec: sequential recommendation with bidirectional encoder representations from transformer. In: *CIKM*, pp. 1441–1450 (2019)
26. Thakoor, S., Tallec, C., Azar, M.G., Munos, R., Veličković, P., Valko, M.: Bootstrapped representation learning on graphs. In: *ICLR Workshop* (2021)
27. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: *International Conference on Learning Representations* (2018)
28. Veličković, P., Fedus, W., Hamilton, W.L., Liò, P., Bengio, Y., Hjelm, R.D.: Deep graph infomax. In: *ICLR* (2018)
29. Wang, K., Zhu, Y., Zang, T., Wang, C., Liu, K., Ma, P.: Multi-aspect graph contrastive learning for review-enhanced recommendation. *ACM TOIS* **42**(2), 1–29 (2023)
30. Wang, M.Y.: Deep graph library: towards efficient and scalable deep learning on graphs. In: *ICLR Workshop* (2019)
31. Wei, C., Liang, J., Bai, B., Liu, D.: Dynamic hypergraph learning for collaborative filtering. In: *CIKM*, pp. 2108–2117 (2022)
32. Wu, C., Wu, F., Qi, T., Ge, S., Huang, Y., Xie, X.: Reviews meet graphs: enhancing user and item representations for recommendation with hierarchical attentive graph neural network. In: *EMNLP-IJCNLP*, pp. 4884–4893 (2019)
33. Wu, J., et al.: Self-supervised graph learning for recommendation. In: *SIGIR*, pp. 726–735 (2021)
34. Wu, Y., Huang, X.: A gumbel-based rating prediction framework for imbalanced recommendation. In: *CIKM*, pp. 2199–2209 (2022)
35. Xia, L., Huang, C., Xu, Y., Zhao, J., Yin, D., Huang, J.: Hypergraph contrastive collaborative filtering. In: *SIGIR*, pp. 70–79 (2022)
36. Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W.L., Leskovec, J.: Graph convolutional neural networks for web-scale recommender systems. In: *SIGKDD*, pp. 974–983 (2018)
37. You, Y., Chen, T., Sui, Y., Chen, T., Wang, Z., Shen, Y.: Graph contrastive learning with augmentations. In: *NeurIPS*, pp. 5812–5823 (2020)
38. Yu, P., Fu, C., Yu, Y., Huang, C., Zhao, Z., Dong, J.: Multiplex heterogeneous graph convolutional network. In: *SIGKDD*, pp. 2377–2387 (2022)

39. Yuan, X., et al.: Multimodal contrastive training for visual representation learning. In: CVPR, pp. 6995–7004 (2021)
40. Zhang, C., Chen, H., Zhang, S., Xu, G., Gao, J.: Geometric inductive matrix completion: a hyperbolic approach with unified message passing. In: WSDM, pp. 1337–1346 (2022)
41. Zhang, J., Shi, X., Zhao, S., King, I.: Star-GCN: stacked and reconstructed graph convolutional networks for recommender systems. In: IJCAI, p. 4264 (2019)
42. Zhang, M., Chen, Y.: Inductive matrix completion based on graph neural networks. In: ICLR (2020)
43. Zhao, X., Zhu, Z., Zhang, Y., Caverlee, J.: Improving the estimation of tail ratings in recommender system with multi-latent representations. In: WSDM, pp. 762–770 (2020)
44. Zhu, Y., Xu, Y., Yu, F., Liu, Q., Wu, S., Wang, L.: Deep graph contrastive representation learning. arXiv preprint [arXiv:2006.04131](https://arxiv.org/abs/2006.04131) (2020)